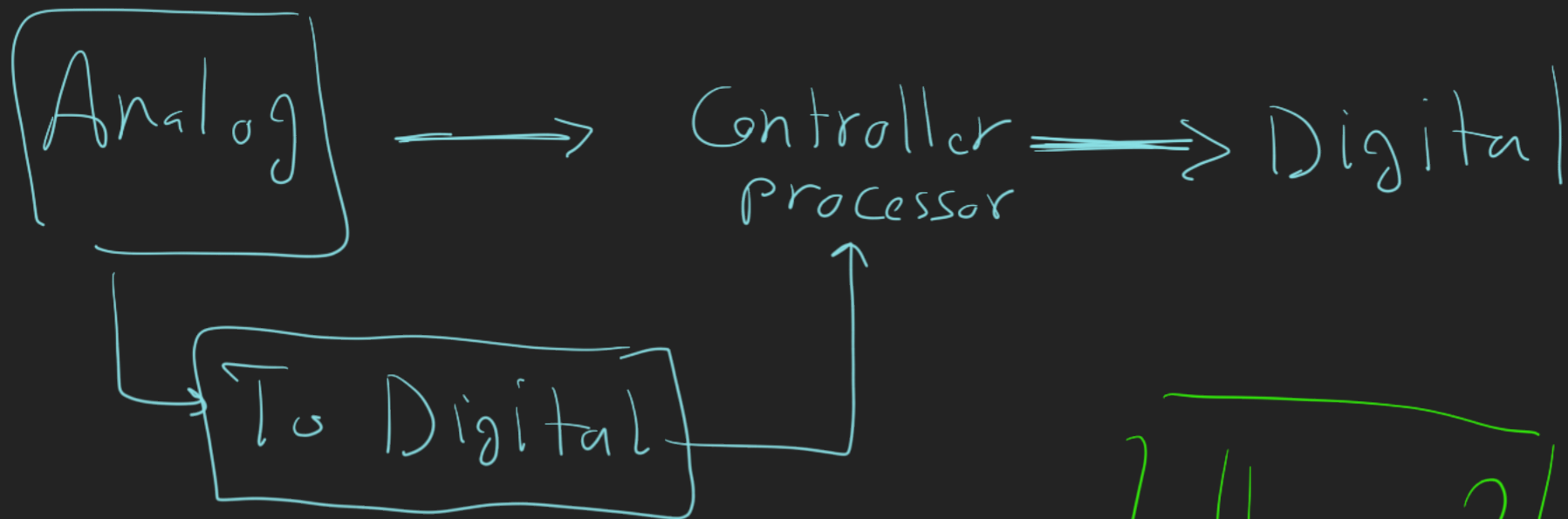
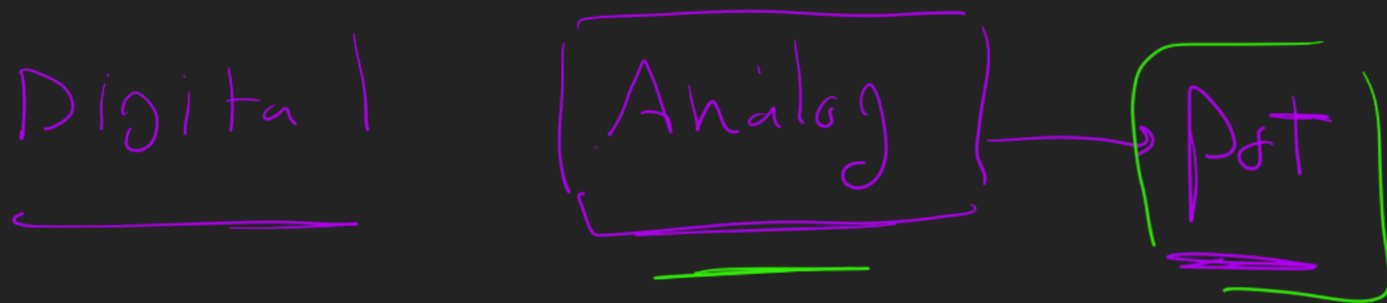
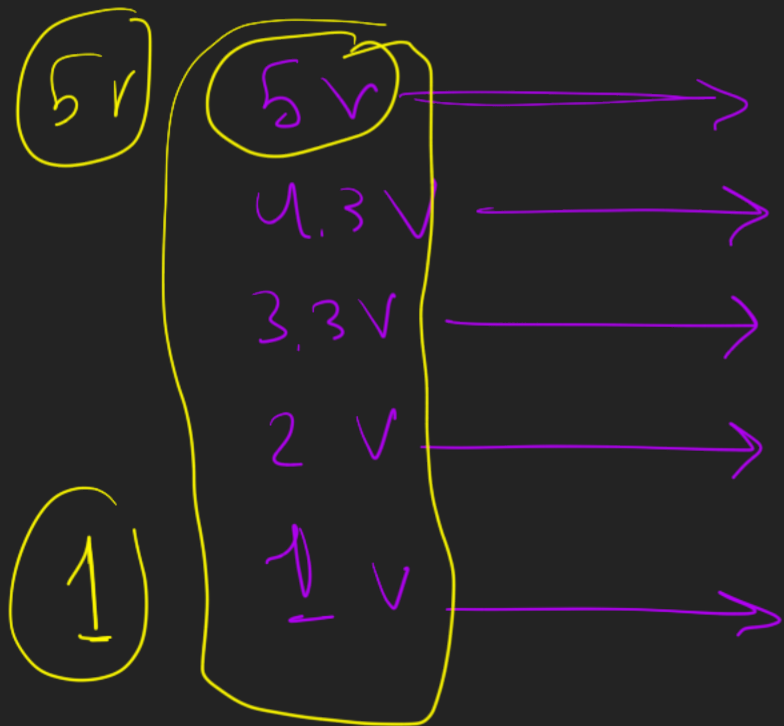


ADC → Analog sensor



[How?]

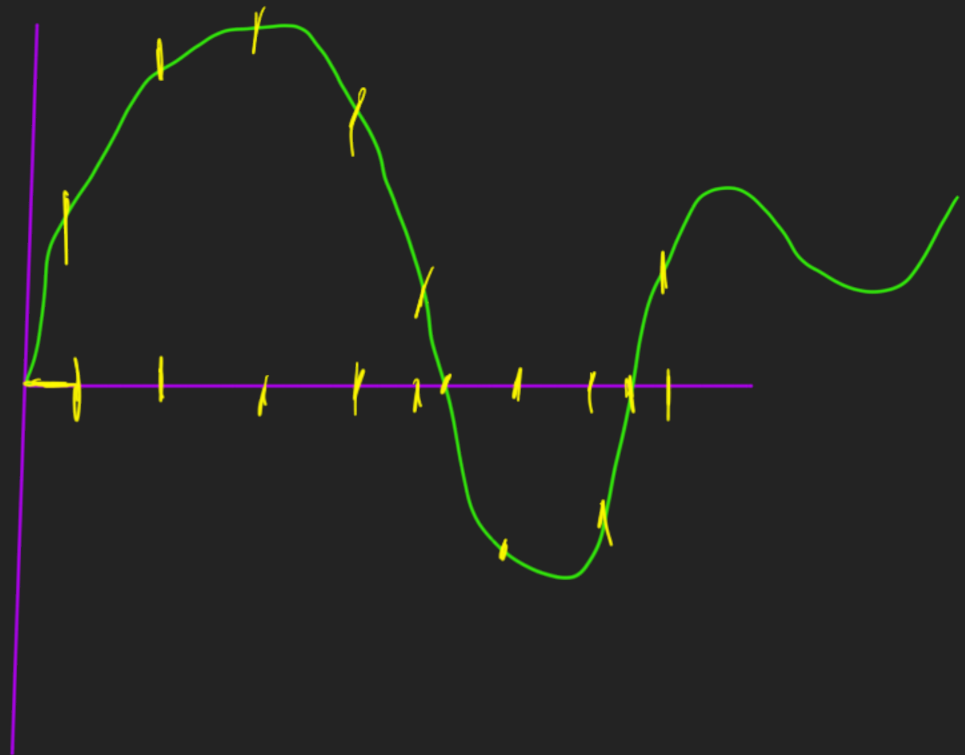
Voltage Division



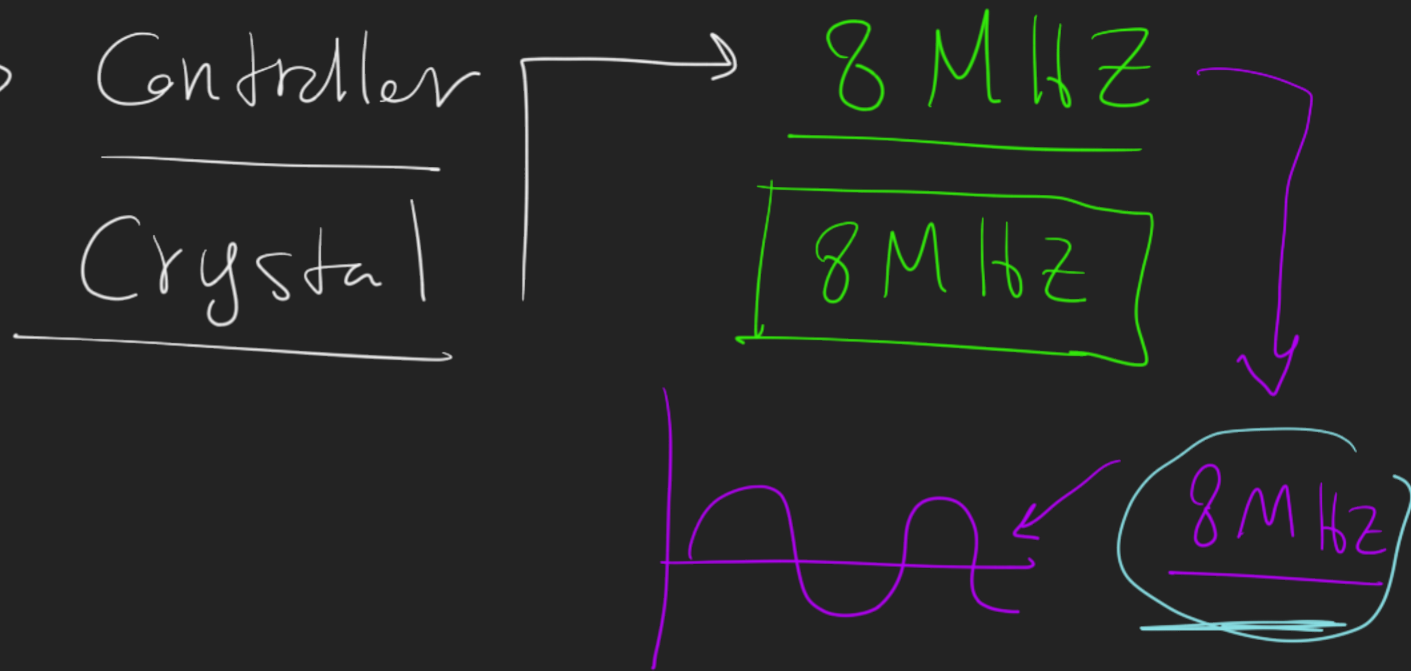
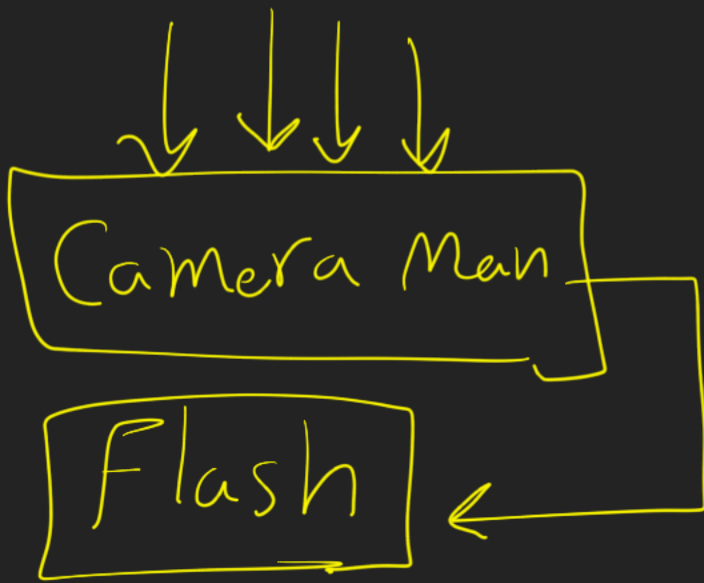
Value, voltage
at a time

5V

Sampling



Frequency $\longrightarrow$ Controller
Crystal



Nyquist



practical
 $\frac{1}{2} f$

$2f$
theoretical

$$f_m = 2 \text{ MHz}$$
$$f_s \geq 4 \text{ MHz}$$

* Digitalization

- ① Sampling ② Quantization

→ Resolution (n) Bits

5v

$$5/4 = \underline{1.25v}$$

step

$$\frac{1024}{256} \rightarrow 10 \text{ Bits}$$
$$\frac{256}{256} \rightarrow 8 \text{ Bits}$$

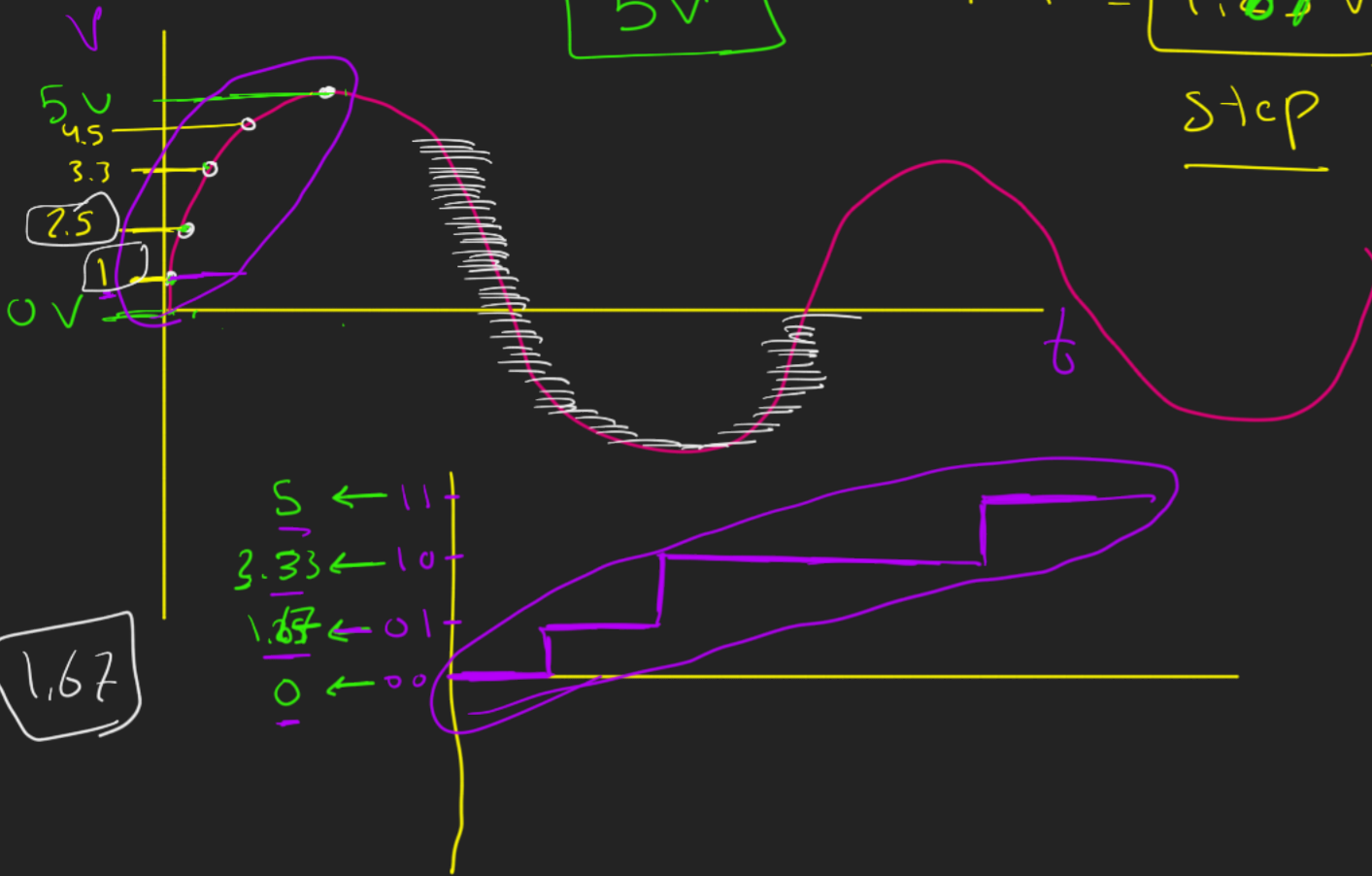
2 Bits

4 Levels

Levels
Bits

$$2^2 \text{ 4 Levels}$$

$$2^{10} = 1024 \text{ Levels}$$



Error

1.5V	→	0
1V	→	0
0.8V	→	0
0.5V	→	0
0V	→	0

Resolution

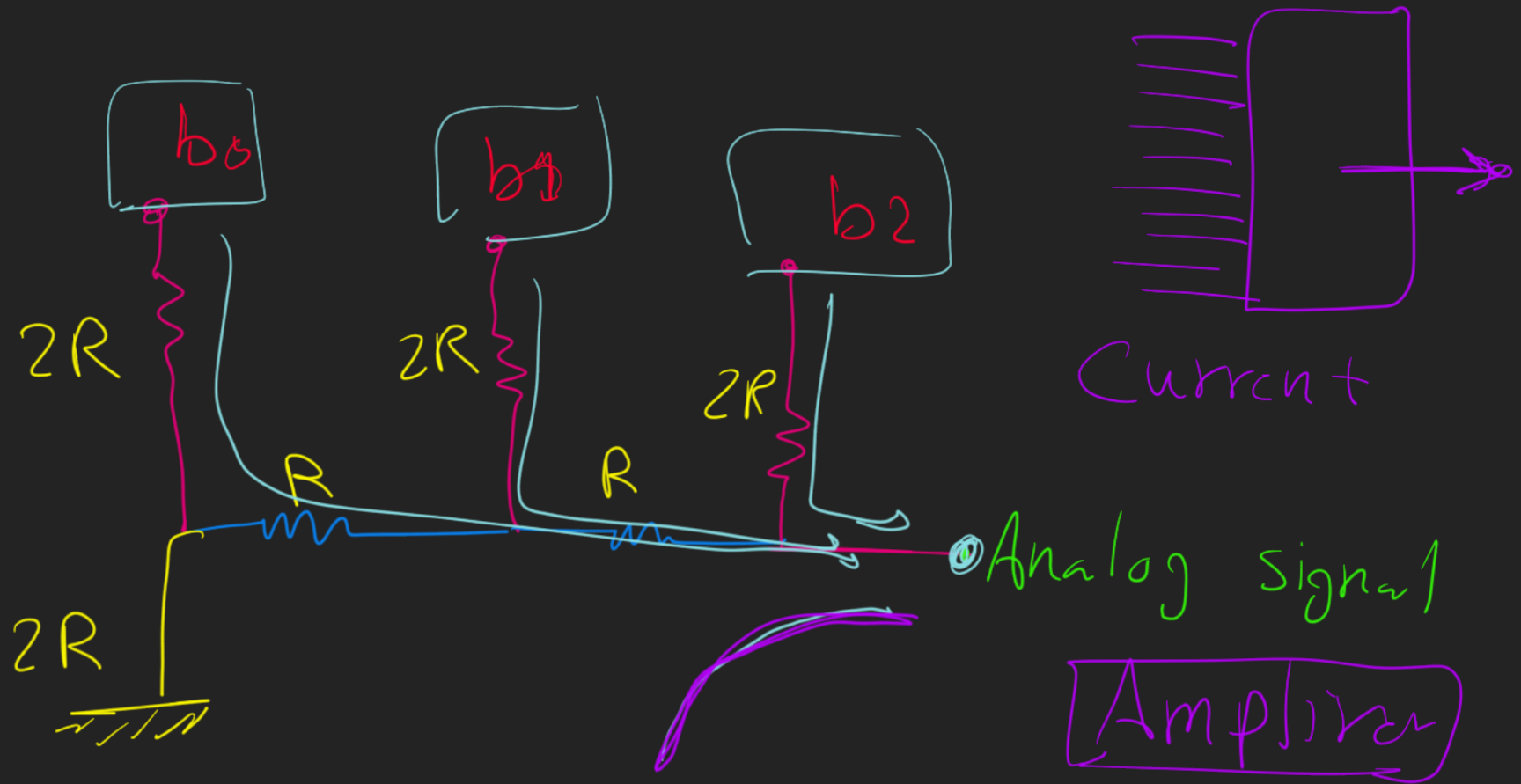
$$\text{Analog value} = \text{Digital value} \times$$

Step size

$$\rightarrow \approx \frac{\text{Max voltage}}{2^n}$$

DAC $\rightarrow$ Digital to Analog

R2R



2) ADC

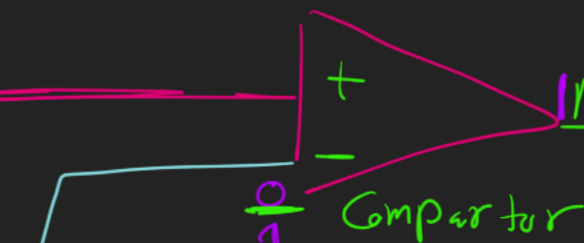
RAMP

SAR

Ramp ADC

I/P

5.3v



CLK

AND

Counter

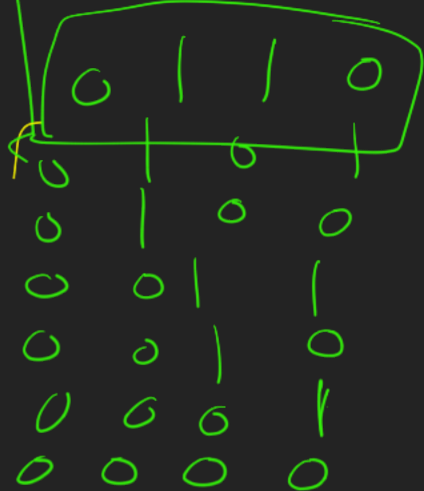
DAC

DAC $\rightarrow$ 4-Bits

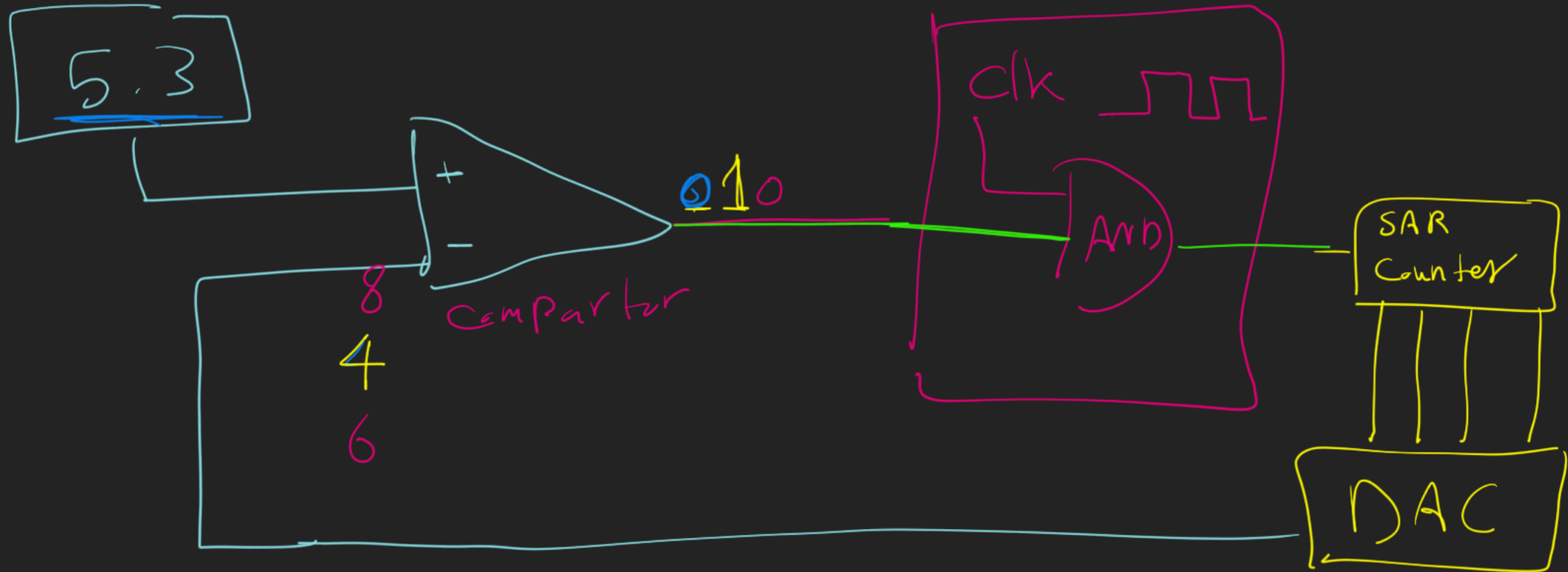
Ref $\rightarrow$ 16 v

$$\text{Step size} = \frac{16}{2^4} = \boxed{1 \text{ v}}$$

$\rightarrow$ Quantization Error



SAR ADC SUCCESSIVE Approximation Register



4 Bits

16v

Step = 1v

5.3 → 5v

<u>b3</u>	b2	b1	b0
(1)	0	0	0
0	(1)	0	0
0	1	1	0
0	1	0	1

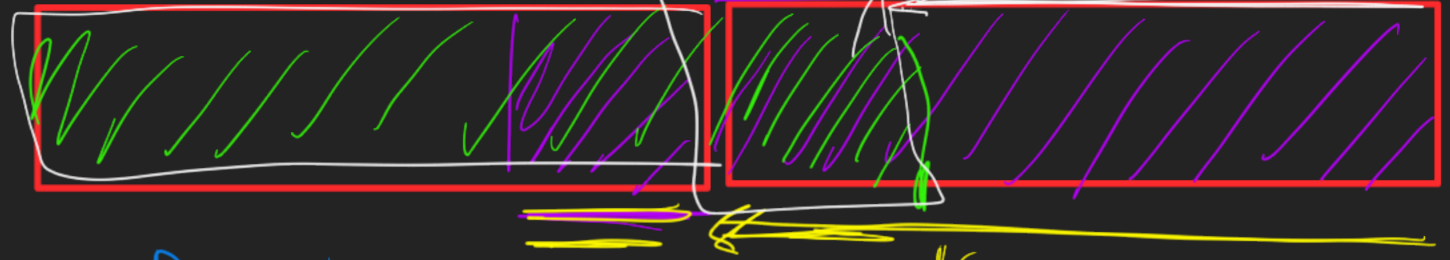
→ (5)v

Atmega 32

ADCH

ADCL

→ Lo-Bits



Registers

Right

ADCL = []

ADCH = (ADCH << 8)

Left

ADCL >> 6

16 Bits

8 Bits

16 Bits

Right

{ U16 { ADCL 0x24
ADCH 0x25
} U16 ADC 0x24 }

* Timers / Counters

↓
Increment

Fixed

+ 1

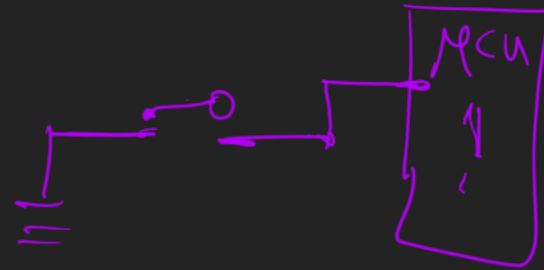
↓
Increment
Not Fixed

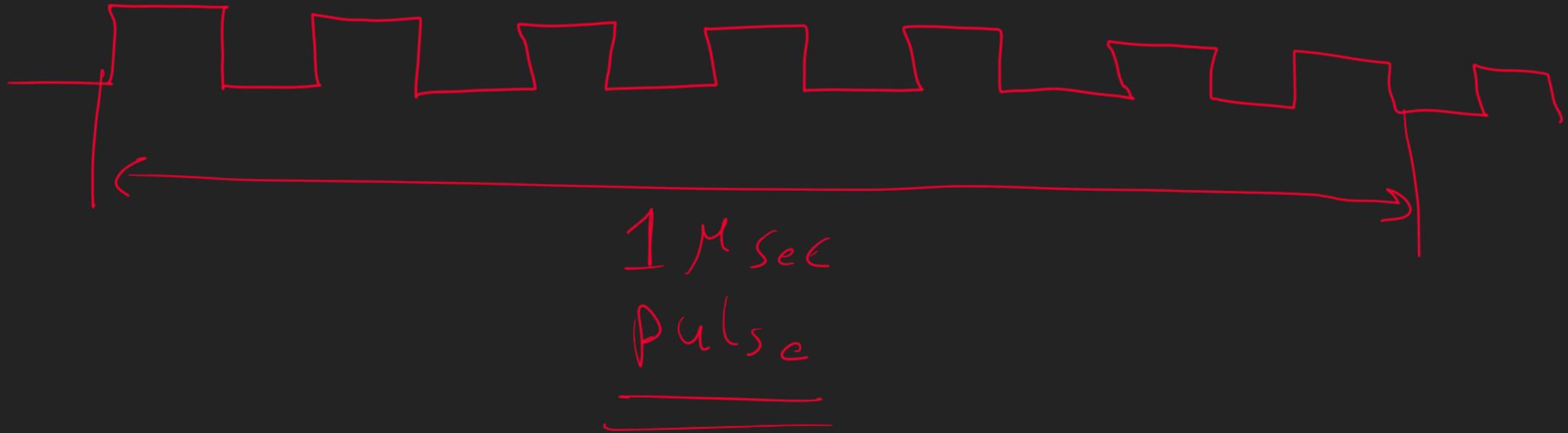
+ 1
+ 2
+ 3

[1] Timer → is a Counter
periodic event



[2] Counter → Not periodic



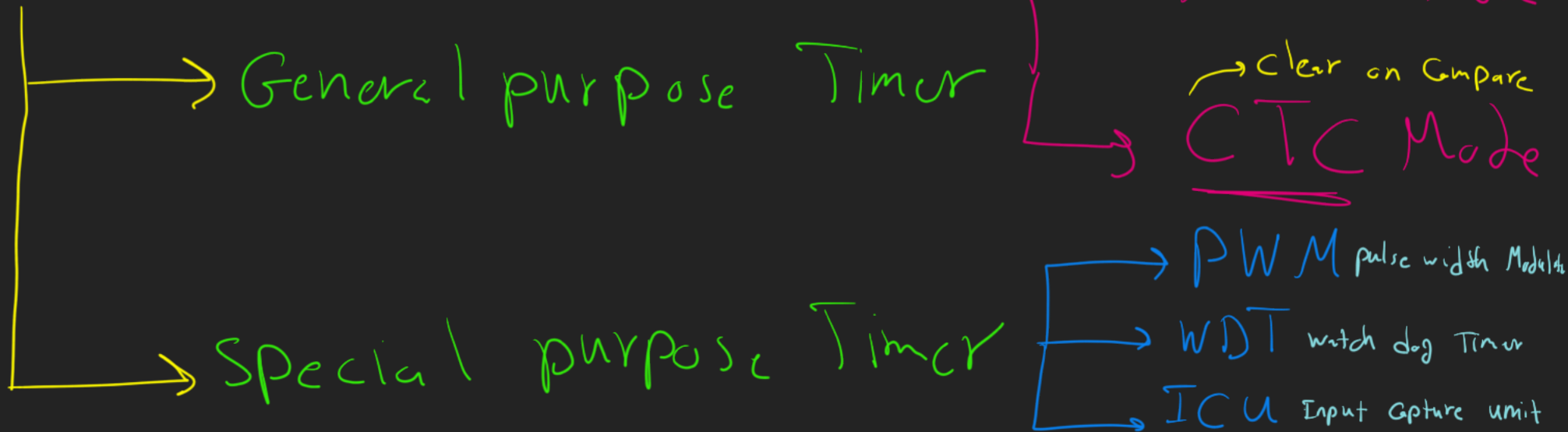


* Resolution (n) $n=8$
 $2^8 = 256$
└─ Bits N_0

* Timer Tick Time = $\frac{1}{\text{Timer freq}}$ = $\frac{1}{1 \text{ MHz}}$ = $\boxed{1 \mu\text{sec}}$

$$\begin{aligned}
 * \text{Timer overflow} &= 2^{\text{Resolution}} \times \text{Tick Time} \\
 &= 256 \times 1 \text{ Msec} = \boxed{256 \text{ Msec}}
 \end{aligned}$$

* Timers

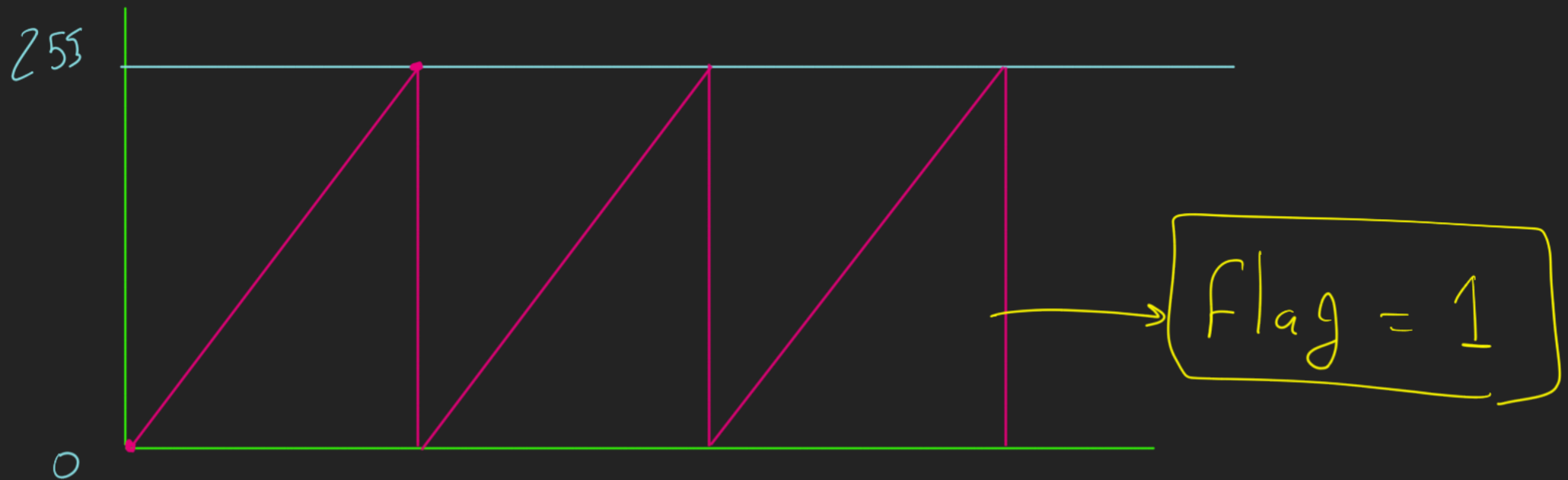


* Normal Mode

set flag Counter $\longrightarrow$ overflow

every 256 Msec

8 Bits



256 Msec

$$ISR(\sigma, \nu f)$$

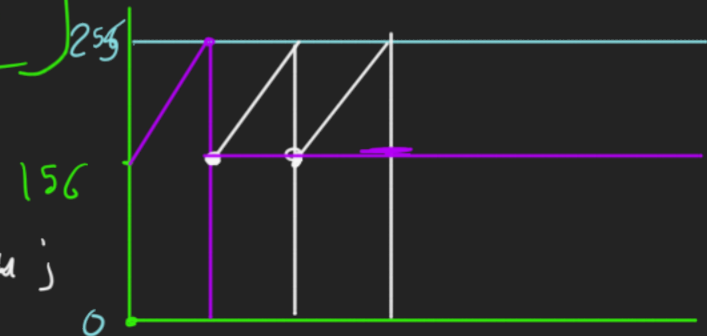
Desired Time = Time over flow ;

Desired Time < Time over flow = 256 μ sec

100 Msec

100 Msec

ISR ()

$$\{ \text{TCNT} = \text{preloadval};$$


Case 3.1

1 sec $\longleftrightarrow$ 1000 msec
1 msec $\longrightarrow$ 1000 μ sec

Desired Time $>$ Time of OV

$$\text{No. of } \overset{\text{overflows}}{\text{OVs}} = \frac{\text{Desired Time}}{\text{Time OV}} = \frac{2.56 \times 1000}{256}$$

ISR (OV) $\xrightarrow{\text{local variable}} = \boxed{10}$

```
{  
  Static U8 Counter = 0;  
  Counter++;  
  If (Counter == 10)  
    Counter = 0;  
}
```

Case 3.2

Desired Time > Overflow Time

$$\text{No. of flow} = \frac{2000}{256} = \boxed{7.8}$$

TickTime = $0.8 \times 256 = \underline{205}$ Tick

$$\text{*preload} = 256 - 205 = \boxed{51}$$

$$\boxed{\text{TCNT} = 51}$$

```
ISR(OV)
{
    static U8 Counter++;
    If (Counter == 8)
        Counter = 0;
    TCNT = 51;
}
```

Timer 10 Bits

$F_{cpu} = 8 \text{ MHz}$

prescaler = 8

Desired Time = 5 sec

- ① Timer Tick Time
- ② Timer freq
- ③ Timer Counter
- ④ Time of overflow
- ⑤ No. of overflows
- ⑥ No. of ticks of flow
- ⑦ preload